Automatic Text-to-Speech System for Vietnamese Using
Outomatiese teks-na-spraak-stelsel vir Viëtnamees gebruik

Artificial Intelligence
Kunsmatige Intelligensie

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Abstract. In the context of digital transformation and the rapid advancement ofartificial intelligence, the demand for Text-to-Speech(TTS) systems hasbecome increasingly critical across domains such as virtual assistants,education, automatic call centers, and accessibility tools for the visuallyimpaired. However, generating natural, context-aware, and prosodicallyappropriate speech remains a significant challenge, particularly forVietnamese—a tonal language with complex pitch contours and regionaldiversity. This research proposes and develops a Vietnamese TTS system builtupon modern deep learning models to overcome the limitations of traditionalmethods. The system leverages state-of-the-art architectures includingTacotron, FastSpeech, and VITS, combined with machine learning techniquesto enhance synthesis quality. Construction and evaluation are performed on adiverse Vietnamese dataset, ensuring adaptability to multiple real-worldcontexts. Results demonstrate that the system achieves natural, clear, highlyflexible speech output with broad potential for deployment in digital platformsand intelligent services.

Abstrak. In die konteks van digitale transformasie en die vinnige vooruitgang van kunsmatige intelligensie, het die vraag na teks-na-spraak (TTS)-stelsels toenemend krities geword oor domeine soos virtuele assistente, onderwys, outomatiese oproepsentrums en toeganklikheidshulpmiddels vir siggestremdes. Die generering van natuurlike, konteksbewuste en prosodies gepaste spraak bly egter 'n beduidende uitdaging, veral vir Viëtnamees - 'n tonale taal met komplekse toonhoogtekontoere en streeksdiversiteit. Hierdie navorsing stel voor en ontwikkel 'n Viëtnamese TTS-stelsel wat gebou is op moderne diepleermodelle om die beperkings van tradisionele metodes te oorkom. Die stelsel maak gebruik van die nuutste argitekture, insluitend Tacotron, FastSpeech en VITS, gekombineer met masjienleertegnieke om sintesekwaliteit te verbeter. Konstruksie en evaluering word op 'n diverse Viëtnamese datastel uitgevoer, wat aanpasbaarheid by verskeie werklike kontekste verseker. Resultate toon dat die stelsel natuurlike, duidelike, hoogs buigsame spraakuitset bereik met breë potensiaal vir ontplooiing in digitale platforms en intelligente dienste.

Keywords: Text-to-Speech (TTS), Artificial Intelligence (AI), Deep Learning,Natural Language Processing (NLP), Speech Synthesis.

Sleutelwoorde: Teks-na-spraak (TTS), Kunsmatige Intelligensie (KI), Deep Learning, Natural Language Processing (NLP), Spraaksintese.

1 Introduction

1 Inleiding

Driven by digital transformation and the progress of artificial intelligence, the need tobuild Text-to-Speech(TTS) systems is becoming ever more important in areas suchas virtual assistants, education, automatic call centers, and support for the visuallyimpaired. The TTS problem, however, extends beyond simply converting text intoaudio; it demands that the synthesized voice be natural, carry appropriate intonation,and fit the given context—a particularly difficult task for Vietnamese, a language witha complex tone system and substantial regional variation (Do, 2026).Traditional concatenative and formant-based TTS methods(Zen et al., 2019) andHidden Markov Model(HMM)-based synthesis(Zen et al., 2019) exhibit limitationsin flexibility, naturalness, and scalability. In response, modern deep learning modelssuch as Tacotron (Wang et al., 2017), Tacotron 2 (Shen et al., 2018), FastSpeech (Ren

Gedryf deur digitale transformasie en die vordering van kunsmatige intelligensie, word die behoefte om teks-na-spraak (TTS)-stelsels te bou al hoe belangriker op gebiede soos virtuele assistente, onderwys, outomatiese oproepsentrums en ondersteuning vir siggestremdes. Die TTS-probleem strek egter verder as die omskakeling van teks in oudio; dit vereis dat die gesintetiseerde stem natuurlik moet wees, gepaste intonasie dra en by die gegewe konteks pas - 'n besonder moeilike taak vir Viëtnamees, 'n taal met 'n komplekse toonstelsel en aansienlike streeksvariasie (Do, 2026). Tradisionele konketenatiewe en formant-gebaseerde TTS-metodes (Zen et al., 2019) en Hidden Markov Model (HMM)-gebaseerde sintese (Zen et al., 2019) toon beperkings in buigsaamheid, natuurlikheid en skaalbaarheid. In reaksie hierop het moderne diepleermodelle soos Tacotron (Wang et al., 2017), Tacotron 2 (Shen et al., 2018), FastSpeech (Ren) et al., 2019,2020), and VITS(Kim et al., 2021) have opened more effective avenues by learning directly from data. Building on these advances, this study focuses on the research and development of a Vietnamese TTS system aimed at improving voice quality and practical applicability.

et al., 2019,2020), en VITS (Kim et al., 2021) het meer effektiewe weë oopgemaak deur direk uit data te leer. Voortbou op hierdie vooruitgang, fokus hierdie studie op die navorsing en ontwikkeling van 'n Viëtnamese TTS-stelsel wat daarop gemik is om stemkwaliteit en praktiese toepaslikheid te verbeter.

2 Related Work

2 Verwante werk

The Vietnamese TTS market features several prominent domestic platforms alongside international solutions. FPT Smart Cloud Text to Speech, developed by FPT Corporation, delivers a high-quality Vietnamese TTS service through a commercial API(Vu et al., 2025; Nguyen, 2023). It applies deep learning architectures such as Tacotron(Wang et al., 2017) and Transformer variants(Vaswani et al., 2017) together with vocoders like WaveNet(Oord et al., 2016) and HiFi-GAN(Kong et al., 2020) to produce natural, region-diverse voices with controllable speed, pitch, and pausing. The system is extensively used in interactive voice response(IVR), audiobooks, virtual assistants, and online education (Do, 2026). Viettel AI Text-to-Speech offers a SaaS-based solution optimized for Vietnamese(Vu et al., 2025; Nguyen, 2023), employing advanced neural models trained on large-scale, Vietnamese-specific speech corpora to ensure accurate pronunciation, natural breaks, and expressive output(Do, 2026; Oord et al., 2016; Kong et al., 2020). Zalo AI Text to Speech, a product of VNG Corporation, integrates deep NLP models like PhoBERT(Nguyen&Tuan, 2020) and BERT-style pre-training(Devlin et al., 2019) with synthesis architectures including Tacotron, FastSpeech, and VITS(Wang et al., 2017; Ren et al., 2019; Kim et al., 2021), leveraging massive user data to support multiple speaking styles(Vu et al., 2025). Voice.vn specializes in speech synthesis and voice cloning(Vu et al., 2025) using end-to-end models like VITS(Kim et al., 2021) and voice conversion techniques(Casanova et al., 2022; Chen et al., 2022), enabling high-fidelity voice cloning and customization for advertising, podcasting, and gaming. Google AI Studio provides a cloud-based environment for TTS via the Gemini family and Google Cloud Text-to-Speech(Vaswani et al., 2017; Ren et al., 2019), featuring multi-lingual, natural, and context-adaptive voice generation (Casanova et al., 2022; Shen et al., 2018). EventLab offers TTS and voice cloning tools built on end-to-end TTS(Wang et al., 2017) and speaker embedding models (Chen et al., 2022; Kim et al., 2021; Casanova et al., 2022). Overall, the market is shifting from simple text-reading systems toward intelligent communication platforms, with strong trends in voice cloning (Jia et al., 2018) and multimodal AI.

Die Viëtnamese TTS-mark beskik oor verskeie prominente plaaslike platforms saam met internasionale oplossings. FPT Smart Cloud Text to Speech, ontwikkel deur FPT Corporation, lewer 'n hoë-gehalte Viëtnamese TTS-diens deur 'n kommersiële API (Vu et al., 2025; Nguyen, 2023). Dit pas diepleer-argitekture soos Tacotron (Wang et al., 2017) en Transformer-variante (Vaswani et al., 2017) toe tesame met vokoders soos WaveNet (Oord et al., 2016) en HiFi-GAN (Kong et al., 2020) om met diverse spoed, stem- en streeksbeheer te produseer. Die stelsel word wyd gebruik in interaktiewe stemrespons (IVR), oudioboeke, virtuele assistente en aanlyn onderwys (Do, 2026). Viettel AI Text-to-Speech bied 'n SaaS-gebaseerde oplossing wat vir Viëtnamees geoptimaliseer is (Vu et al., 2025; Nguyen, 2023), wat gebruik maak van gevorderde neurale modelle wat op grootskaalse, Viëtnamees-spesifieke spraakkorpora opgelei is om akkurate 206-uitspraak, en ekspressiewe, natuurlike uitspraak, oor- en uitspraak te verseker. al., 2016; Kong et al., 2020). Zalo AI Text to Speech, 'n produk van VNG Corporation, integreer diep NLP-modelle soos PhoBERT(Nguyen&Tuan, 2020) en BERT-styl vooropleiding (Devlin et al., 2019) met sintese-argitekture insluitend Tacotron, FastSpeech, en Rens et al., VITS, 20 et al. 2019; Kim et al., 2021), wat massiewe gebruikersdata gebruik om veelvuldige praatstyle te ondersteun (Vu et al., 2025). Voice.vn spesialiseer in spraaksintese en stemkloning (Vu et al., 2025) deur gebruik te maak van end-tot-end-modelle soos VITS (Kim et al., 2021) en stemomskakelingstegnieke (Casanova et al., 2022; Chen et al., 2022), wat pasmaak vir stemafsluiting moontlik maak en pasmaak vir advertensies. podcasting en speletjies. Google AI Studio verskaf 'n wolk-gebaseerde omgewing vir TTS via die Gemini-familie en Google Cloud Text-to-Speech (Vaswani et al., 2017; Ren et al., 2019), met veeltalige, natuurlike en konteks-aanpasbare stemgenerering (She et al., 20182., 20182). EventLab bied TTS- en stemkloningnutsmiddels gebou op end-tot-end TTS (Wang et al., 2017) en luidsprekerinbeddingsmodelle (Chen et al., 2022; Kim et al., 2021; Casanova et al., 2022). In die algemeen skuif die mark van eenvoudige teksleesstelsels na intelligente kommunikasieplatforms, met sterk neigings in stemkloning (Jia et al., 2018) en multimodale KI.

The open-source community has contributed a range of Vietnamese TTS models. VietTTS is a representative system built with encoder-decoder or Transformer architectures(Wang et al., 2017; Vaswani et al., 2017; Ren et al., 2019), trained on carefully normalized Vietnamese data(Do, 2026) to handle tones, compound words, and sentence structures, offering near-real-time latency suitable for virtual assistants, e-learning, and audiobook production(Nguyen, 2023; Vu et al., 2025). NeuTTS-Air focuses on lightweight, efficient inference, balancing voice quality and speed to operate on CPUs or low-resource devices(Ren et al., 2019,2020; Kim et al., 2021),

Die oopbrongemeenskap het 'n reeks Viëtnamese TTS-modelle bygedra. VietTTS is 'n verteenwoordigende stelsel wat gebou is met enkodeerder-dekodeerder of transformator-argitekture (Wang et al., 2017; Vaswani et al., 2017; Ren et al., 2019), opgelei op noukeurig genormaliseerde Viëtnamese data (Do, 2026) om tone, saamgestelde woorde en sintydstrukture te hanteer, wat geskikte naby-tyd-strukture bied, wat werklike latensie bied. e-leer en oudioboekproduksie (Nguyen, 2023; Vu et al., 2025). NeuTTS-Air fokus op liggewig, doeltreffende afleiding, balansering van stemkwaliteit en spoed om op SVE's of laehulpbrontoestelle te werk (Ren et al., 2019,2020; Kim et al., 2021), optimized for chatbots, IVR, and IoT applications (Ping et al., 2017; Kalchbrenner et al., 2018). Kani-TTS is a large-scale model with approximately 370 million parameters, aiming for high expressiveness and professional audio quality (Kim et al., 2021; Casanova et al., 2022), excelling in audiobook and media production but requiring powerful GPUs for deployment(Ren et al., 2019; Ping et al., 2017; Kalchbrenner et al., 2018). VieNeu-TTS combines TTS with voice cloning, extracting speaker characteristics via techniques like AutoVC(Qian et al., 2019) and StarGAN-VC(Kameoka et al., 2018) and applying them to synthesized speech (Casanova et al., 2022; Jia et al., 2018), enabling strong personalization but also raising ethical and security concerns. ViXTTS serves as a demonstration project for voice cloning, illustrating the technology for rapid experimentation(Thinh, 2024; Kim et al., 2021). Valtec-TTS is a compact model optimized for extremely resource-constrained environments such as embedded devices and IoT(Ren et al., 2019,2020; Ping et al., 2017; Kalchbrenner et al., 2018), delivering adequate clarity for simple notification and guidance applications. Collectively, these models illustrate the maturation of Vietnamese TTS, with specialized branches ranging from high-quality synthesis to lightweight deployment and voice personalization.

Bydraetitel (verkort indien te lank) 3 geoptimaliseer vir chatbots, IVR en IoT-toepassings (Ping et al., 2017; Kalchbrenner et al., 2018). Kani-TTS is 'n grootskaalse model met ongeveer 370 miljoen parameters, wat mik na hoë ekspressiwiteit en professionele klankgehalte (Kim et al., 2021; Casanova et al., 2022), wat uitblink in oudioboek- en mediaproduksie, maar kragtige GPU's benodig vir ontplooiing (Ren et al., 2019; Kall et al., 2019 et al. 2018). VieNeu-TTS kombineer TTS met stemkloning, onttrek sprekerkenmerke via tegnieke soos AutoVC (Qian et al., 2019) en StarGAN-VC (Kameoka et al., 2018) en pas dit toe op gesintetiseerde spraak (Casanova et al., 2022; Jiabling et al., 2018, maar ook personalization), maar ook. etiese en veiligheidskwessies. ViXTTS dien as 'n demonstrasieprojek vir stemkloning, wat die tegnologie vir vinnige eksperimentering illustreer (Thinh, 2024; Kim et al., 2021). Valtec-TTS is 'n kompakte model wat geoptimaliseer is vir uiters hulpbronbeperkte omgewings soos ingebedde toestelle en IoT (Ren et al., 2019,2020; Ping et al., 2017; Kalchbrenner et al., 2018), wat voldoende duidelikheid lewer vir eenvoudige toepassingkennisgewing en leiding. Gesamentlik illustreer hierdie modelle die rypwording van Viëtnamese TTS, met gespesialiseerde takke wat wissel van hoëgehalte-sintese tot liggewig-ontplooiing en stemverpersoonliking.

Data quality is foundational for TTS. The VLSP Speech Corpus is a large, acadeically oriented dataset with multi-speaker recordings in controlled conditions (Ardila et al., 2020; Vu et al., 2025), richly annotated with phonetic and regional information(Yamagishi et al., 2019; Do, 2026). VIVOS is a smaller open-source corpus (around 10–20 hours) of clean read speech (Ardila et al., 2020; Ito & Johnson, 2017), widely used for initial modeling despite limited emotional and real-world diversity(Do, 2026). Commercial datasets from FPT Corporation, Viettel Group, and VNG Corporation are far larger and more detailed. FPT's dataset comprises hundreds to thousands of hours of studio-quality recordings with exhaustive annotations (Vu et al., 2025; Do, 2026; Oord et al., 2016; Kong et al., 2020). Viettel's data blends studio and real-world recordings covering diverse accents (Vu et al., 2025; Yamagishi et al., 2019; Jia et al., 2018; Do, 2026). Zalo's data benefits from its broad user ecosystem, incorporating contemporary language use(Vu et al., 2025; Nguyen&Tuan, 2020; Devlin et al., 2019; Kim et al., 2021). Additionally, VietTTS provides a basic open dataset with text–audio pairs sufficient for small-scale experiments(Nguyen, 2023;

Ito&Johnson, 2017; Vu et al., 2025). Overall, dataset scale, diversity, annotation depth, and realism directly determine TTS performance.

Datakwaliteit is die grondslag vir TTS. Die VLSP Speech Corpus is 'n groot, akademies georiënteerde datastel met multi-luidspreker opnames in gekontroleerde toestande (Ardila et al., 2020; Vu et al., 2025), ryklik geannoteer met fonetiese en streeksinligting (Yamagishi et al., 2019; Do, 2026). VIVOS is 'n kleiner oopbronkorpus (ongeveer 10–20 uur) van skoongeleesde spraak (Ardila et al., 2020; Ito & Johnson, 2017), wat wyd gebruik word vir aanvanklike modellering ondanks beperkte emosionele en werklike diversiteit (Do, 2026). Kommersiële datastelle van FPT Corporation, Viettel Group en VNG Corporation is baie groter en meer gedetailleerd. FPT se datastel bestaan uit honderde tot duisende ure se opnames van ateljeegehalte met volledige aantekeninge (Vu et al., 2025; Do, 2026; Oord et al., 2016; Kong et al., 2020). Viettel se data kombineer ateljee- en werklike wêreldopnames wat uiteenlopende aksente dek (Vu et al., 2025; Yamagishi et al., 2019; Jia et al., 2018; Do, 2026). Zalo se data trek voordeel uit sy breë gebruikersekosisteem, wat kontemporêre taalgebruik insluit (Vu et al., 2025; Nguyen&Tuan, 2020; Devlin et al., 2019; Kim et al., 2021). Daarbenewens bied VietTTS 'n basiese oop datastel met teks-klankpare wat voldoende is vir kleinskaalse eksperimente (Nguyen, 2023; Ito&Johnson, 2017; Vu et al., 2025). In die algemeen bepaal datastelskaal, diversiteit, annotasiediepte en realisme TTS-prestasie direk.

3 System Design

3 Stelselontwerp

Early TTS systems relied on concatenative and formant-based methods, which assemble or mathematically model speech units but suffer from inflexibility, high database costs, and unnatural prosody(Zen et al., 2019). Statistical parametric synthesis using Hidden Markov Models(HMM) improved flexibility but still produced monotonous, less natural output(Zen et al., 2019). The advent of deep learning brought end-to-end models such as Tacotron and Tacotron 2, which employ encoder–decoder architectures with attention to directly map text to

Vroeë TTS-stelsels het staatgemaak op aaneenlopende en formant-gebaseerde metodes, wat spraakeenhede saamstel of wiskundig modelleer, maar ly aan onbuigsaamheid, hoë databasiskoste en onnatuurlike prosodie (Zen et al., 2019). Statistiese parametriese sintese met behulp van Hidden Markov Models (HMM) het buigsaamheid verbeter, maar steeds eentonige, minder natuurlike uitset gelewer (Zen et al., 2019). Die koms van diep leer het end-tot-end-modelle gebring soos Tacotron en Tacotron 2, wat enkodeerder-dekodeerder-argitekture gebruik met aandag aan direk kaartteks na mel-spectrograms, followed by vocoders like WaveNet(Oord et al., 2016) or WaveGlow(Prenger et al., 2019) to generate waveforms(Wang et al., 2017; Shen et al., 2018; Vaswani et al., 2017). These models automatically learn phonetic and prosodic features, yielding significantly more natural speech, but have limitations in inference speed and attention errors. FastSpeech and FastSpeech 2 addressed these issues by removing attention during inference and using duration, pitch, and energy predictors, achieving faster and more stable synthesis with explicit prosody control (Ren et al., 2019,2020; Vaswani et al., 2017). The latest leap, VITS, integrates a variational autoencoder(Kingma&Welling, 2014) with a generative adversarial network(Goodfellow et al., 2014) to directly generate waveforms from text in a single end-to-end framework, eliminating a separate vocoder and further boosting quality and efficiency(Kim et al., 2021; Casanova et al., 2022). Voice cloning and multi-speaker synthesis have also advanced, using speaker embeddings and style tokens to enable personalization and zero-shot adaptation(Jia et al., 2018; Casanova et al., 2022; Chen et al., 2022; Wang et al., 2018). Additionally, transfer and self-supervised learning methods(Baevski et al., 2020; Hsu et al., 2021; Radford et al., 2022) mitigate data scarcity for Vietnamese by pretraining on large multilingual corpora. A modern TTS pipeline typically includes data collection and preprocessing, text normalization (Do, 2026), model training, and waveform generation (Wang et al., 2017; Kim et al., 2021).

mel-spektrogramme, gevolg deur vokoders soos WaveNet (Oord et al., 2016) of WaveGlow (Prenger et al., 2019) om golfvorms te genereer (Wang et al., 2017; Shen et al., 2018; Vaswani et al., 2017). Hierdie modelle leer outomaties fonetiese en prosodiese kenmerke, wat aansienlik meer natuurlike spraak lewer, maar het beperkings in afleidingsspoed en aandagfoute. FastSpeech en FastSpeech 2 het hierdie kwessies aangespreek deur aandag tydens afleiding te verwyder en duur-, toonhoogte- en energievoorspellers te gebruik, om vinniger en meer stabiele sintese met eksplisiete prosodiebeheer te bewerkstellig (Ren et al., 2019,2020; Vaswani et al., 2017). Die jongste sprong, VITS, integreer 'n variasie-outo-enkodeerder (Kingma&Welling, 2014) met 'n generatiewe teenstandernetwerk (Goodfellow et al., 2014) om golfvorms direk vanaf teks in 'n enkele end-tot-end-raamwerk te genereer, wat 'n aparte vocoder uitskakel en gehalte en doeltreffendheid verder bevorder., nova20 et al. 2022). Stemkloning en multi-luidsprekersintese het ook gevorder, met behulp van luidsprekerinbeddings en styltekens om verpersoonliking en nulskoot-aanpassing moontlik te maak (Jia et al., 2018; Casanova et al., 2022; Chen et al., 2022; Wang et al., 2018). Boonop versag oordrag en self-toesig leermetodes (Baevski et al., 2020; Hsu et al., 2021; Radford et al., 2022) dataskaarste vir Viëtnamees deur vooropleiding op groot veeltalige korpus. ’n Moderne TTS-pyplyn sluit tipies data-insameling en voorverwerking, teksnormalisering (Do, 2026), modelopleiding en golfvormgenerering in (Wang et al., 2017; Kim et al., 2021).

The proposed system implements an emotional TTS pipeline: input text containing emotion tags is segmented by emotion, each segment is synthesized via viXTTS, and the resulting waveforms are concatenated with silence gaps to produce the final audio file

Die voorgestelde stelsel implementeer 'n emosionele TTS-pyplyn: invoerteks wat emosie-etikette bevat, word volgens emosie gesegmenteer, elke segment word via viXTTS gesintetiseer, en die gevolglike golfvorms word aaneengeskakel met stiltegapings om die finale oudiolêer te produseer

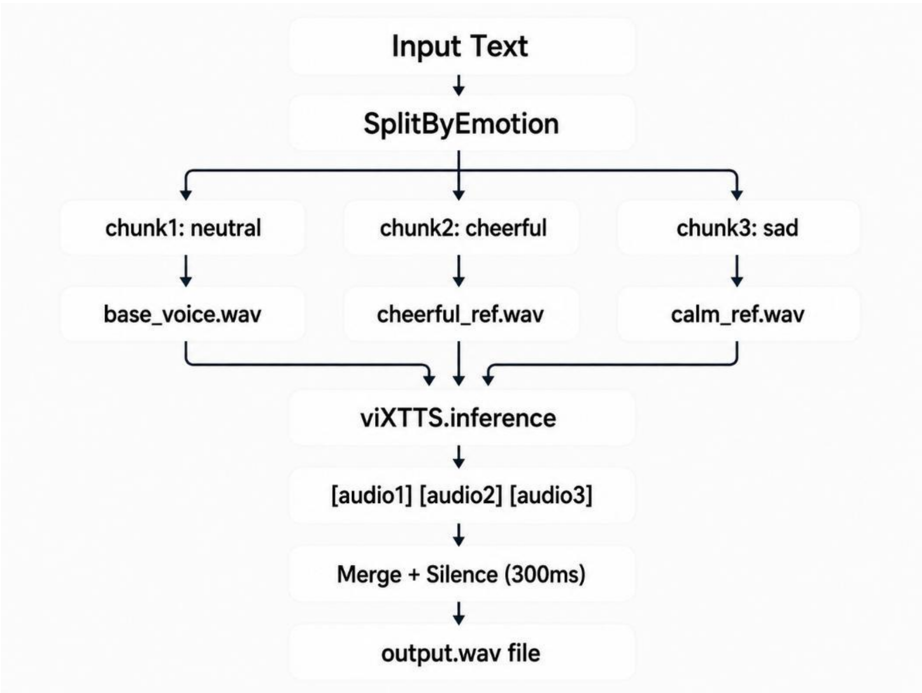

Fig. 1. *Workflow of the Proposed Emotion-Based Speech Synthesis System*

Fig. *1. Werkvloei van die voorgestelde emosie-gebaseerde spraaksintesestelsel*

In the first layer, text is scanned line by line. When a line contains an emotion tag in parentheses, the system maps it to an emotion label using a Vietnamese keyword dictionary. Consecutive lines sharing the same emotion are grouped into a chunk of (text, emotion). Before synthesis, the text is cleaned by removing tag content, normalizing spaces, and standardizing Vietnamese orthography. In the second layer, each chunk is passed to the viXTTS inference engine together with speaker characteristics extracted from a reference audio file. An emotion map assigns a

In die eerste laag word teks reël vir reël geskandeer. Wanneer 'n reël 'n emosie-etiket tussen hakies bevat, karteer die stelsel dit na 'n emosie-etiket deur 'n Viëtnamese sleutelwoordwoordeboek te gebruik. Opeenvolgende reëls wat dieselfde emosie deel, word in 'n stuk (teks, emosie) gegroepeer. Voor sintese word die teks skoongemaak deur merker-inhoud te verwyder, spasies te normaliseer en Viëtnamese ortografie te standaardiseer. In die tweede laag word elke deel na die viXTTS-inferensie-enjin oorgedra tesame met luidsprekerkenmerke wat uit 'n verwysingsoudiolêer onttrek is. 'n Emosiekaart ken 'n toe

In die eerste laag woord teksreël vir reële geskandeer. Wanneer 'n reël 'n emosie-etiket tussen hakies bevat, kaart die stelsel dit na 'n emosie-etiket deur 'n Viëtnamese sleutelwoordeboek te gebruik. Openvolgende reëls wat dieselfde emosie deel, word in 'n stuk (teks, emosie) gegroepeer. Vir sintese word die teks skoon merker-inhoud te verwyder, spasies te normaliseer in Vietnamese ortografie te standaardiseer. In die tweede laag word elke deel na die viXTTS-inferensie-enjin oorgedra tesame met luidsprekermerke wat uit 'n verwysingsoudiolêer onttrek is. 'n Emosiekaart ken 'n toe

In die eerste laag woord teksreël vir reële geskandeer. Wanneer 'n reël 'n emosie-etiket tussen hakies bevat, word die stelsel dit na 'n emosie-kaart-etiket deur 'n Viëtnamese sleutelwoordeboek te gebruik. Openvolgende reëls wat dieselfde emosie deel, word in 'n stuk (teks, emosie) gegroepeer. Vir sintese word die teks skoon merker-inhoud te verwyder, spasies te normaliseer in Vietnamese ortografie te standaardiseer. In die tweede laag word elke deel na die viXTTS-inferensie-enjin oorgedra tesame met luidsprekermerke wat uit 'n verwysingsoudiolêer onttrek is. 'n Emosiekaart ken 'n toe

reference file per emotion (e.g., neutral → base_voice.wav, cheerful →
verwysing lêer emosie (bv. neutraal vrolik

cheerful_ref.wav). To maintain consistent speaker identity, latent speaker embeddings are extracted once from a user-provided voice sample and reused across all chunks (Jia et al., 2018; Casanova et al., 2022). The synthesis produces a waveform for each

cheerful_ref.wav). Om konsekwente spreker-identiteit te handhaaf, word latente spreker-inbeddings een keer uit 'n gebruiker-verskafde stemmonster onttrek en oor alle stukke hergebruik (Jia et al., 2018; Casanova et al., 2022). Die sintese produseer 'n golfvorm vir elkeen

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[redacted]

chunk, as formalized in the following algorithm

stuk, soos geformaliseer in die volgende algoritme

In the third layer, the generated waveforms are concatenated sequentially with 300 ms of silence inserted to separate emotional turns and avoid abrupt cuts. The final output is saved as a WAV file. This design effectively combines keyword-based emotion classification with conditional TTS and audio post-processing, enabling tag-driven, expressive speech synthesis using a unified voice identity.

In die derde laag word die gegenereerde golfvorms opeenvolgend aaneengeskakel met 300 ms se stilte wat ingevoeg word om emosionele draaie te skei en abrupte snye te vermy. Die finale uitset word as 'n WAV-lêer gestoor. Hierdie ontwerp kombineer effektief sleutelwoord-gebaseerde emosieklassifikasie met voorwaardelike TTS en oudio-naverwerking, wat merkergedrewe, ekspressiewe spraaksintese moontlik maak deur 'n verenigde stemidentiteit te gebruik.

4 Evaluation Metrics

4 Evalueringsmetrieke

To comprehensively assess the text-to-speech and voice cloning system, two complementary deep-learning-based metrics are employed: speaker similarity via Resemblyzer, and perceived speech quality via DNSMOS.

Om die teks-na-spraak- en stemkloningstelsel volledig te assesseer, word twee komplementêre diep-leer-gebaseerde maatstawwe gebruik: sprekerooreenkoms via Resemblyzer, en waargenome spraakkwaliteit via DNSMOS.

Resemblyzer is a neural network-based tool for speaker verification and voice similarity measurement(Mishra et al., 2023). At its core, it employs a speaker verification encoder trained with a generalized end-to-end loss (Wan et al., 2018). The operational principle is as follows: first, an input utterance—either the original reference voice or the synthesized cloned voice—is processed by a deep neural network that extracts a fixed-dimensional embedding vector(often referred to as a d-vector or speaker embedding). This vector is designed to capture the unique vocal identity of the speaker, including characteristics such as timbre, pitch contour, and speaking style, while being largely invariant to the linguistic content or the specific words spoken. In other words, regardless of what the person says, the embedding should remain consistent for the same speaker and distinctly different for different speakers (Wan et al., 2018).

Resemblyzer is 'n neurale netwerk-gebaseerde instrument vir sprekerverifikasie en stemooreenkomsmeting (Mishra et al., 2023). In sy kern gebruik dit 'n sprekerverifikasie-enkodeerder wat opgelei is met 'n algemene end-tot-end verlies (Wan et al., 2018). Die operasionele beginsel is soos volg: eerstens, 'n insetuiting - óf die oorspronklike verwysingstem óf die gesintetiseerde gekloonde stem - word verwerk deur 'n diep neurale netwerk wat 'n vaste-dimensionele inbeddingsvektor onttrek (dikwels na verwys as 'n d-vektor of luidsprekerinbedding). Hierdie vektor is ontwerp om die unieke vokale identiteit van die spreker vas te vang, insluitend eienskappe soos timbre, toonhoogte kontoer en praatstyl, terwyl dit grootliks onveranderlik is van die linguistiese inhoud of die spesifieke woorde wat gespreek word. Met ander woorde, ongeag wat die persoon sê, moet die inbedding konsekwent bly vir dieselfde spreker en duidelik verskillend vir verskillende sprekers (Wan et al., 2018).

Once the embedding vectors are obtained for both the reference audio A(the original speaker's voice) and the synthesized audio B(the cloned voice), their similarity is quantified using the cosine similarity metric:

Sodra die inbeddingsvektore vir beide die verwysingsoudio A (die oorspronklike spreker se stem) en die gesintetiseerde oudio B (die gekloonde stem) verkry is, word hul ooreenkoms gekwantifiseer deur die cosinus-ooreenkoms metriek te gebruik:

d

C osineSimilarity A, B= A ·B =σ i=1 A i B i

C osineGelykvormigheid A, B= A ·B =σ i=1 A i B i

A Bσd

'n Bσd

i=1 A i 2 σ d

i=1 A i 2 σ d

i=1 B i 2

i=1 B i 2

The cosine similarity measures the angle between the two vectors in the high-dimensional embedding space. A value close to 1 indicates that the vectors point in nearly the same direction, meaning the two voices are perceived as highly similar or almost identical in terms of speaker identity(Jia et al., 2018; Wan et al., 2018). Conversely, a value close to 0 or negative would suggest dissimilar speakers. Thus, Resemblyzer provides an objective, automatic, and reproducible metric for evaluating how faithfully a voice cloning system preserves the original speaker's vocal characteristics.

Die cosinus-ooreenkoms meet die hoek tussen die twee vektore in die hoë-dimensionele inbeddingsruimte. 'n Waarde naby aan 1 dui aan dat die vektore in byna dieselfde rigting wys, wat beteken dat die twee stemme as hoogs eenders of amper identies in terme van sprekeridentiteit beskou word (Jia et al., 2018; Wan et al., 2018). Omgekeerd, 'n waarde naby aan 0 of negatief sal verskillende sprekers voorstel. Resemblyzer verskaf dus 'n objektiewe, outomatiese en reproduceerbare maatstaf om te evalueer hoe getrou 'n stemkloningstelsel die oorspronklike spreker se vokale eienskappe bewaar.

DNSMOS(Deep Noise Suppression Mean Opinion Score) is a non-intrusive, deep learning–based metric designed to estimate the perceptual quality of speech signals without requiring a clean reference audio signal(Reddy et al., 2021). This makes it

DNSMOS (Deep Noise Suppression Mean Opinion Score) is 'n nie-indringende, diep leer-gebaseerde maatstaf wat ontwerp is om die perseptuele kwaliteit van spraakseine te skat sonder om 'n skoon verwysing oudiosein te vereis (Reddy et al., 2021). Dit maak dit

5 Experiments and Results
5 Eksperimente en resultate

Training Dataset The system was trained and evaluated using the viVoice corpus (Le, 2024), the first publicly available large-scale Vietnamese speech dataset designed for multi-speaker text-to-speech. viVoice comprises over 1,000 hours of cleaned audio and accompanying transcriptions sourced from 186 YouTube channels. All audio samples are provided at a 24 kHz sampling rate in WAV format, with background music and noise removed to ensure clean training data. The dataset covers a wide variety of speakers, accents (Northern, Central, Southern), and speaking styles, making it particularly suitable for training robust multi-speaker Vietnamese TTS models such as XTTS-v2. viVoice is released under the CC BY-NC-SA 4.0 license.
OpleidingsdatastelDie stelsel is opgelei en geëvalueer deur gebruik te maak van die viVoice-korpus (Le, 2024), die eerste publiek-beskikbare grootskaalse Viëtnamese spraakdatastel wat ontwerp is vir veelsprekers-teks-na-spraak. viVoice bestaan uit meer as 1 000 uur se skoongemaakte oudio en gepaardgaande transkripsies afkomstig van 186 YouTube-kanale. Alle oudiomonsters word verskaf teen 'n 24 kHz-steekproeftempo in WAV-formaat, met agtergrondmusiek en geraas verwyder om skoon opleidingsdata te verseker. Die datastel dek 'n wye verskeidenheid luidsprekers, aksente (Noordelike, Sentraal, Suidelike) en praatstyle, wat dit veral geskik maak vir die opleiding van robuuste multi-luidspreker Viëtnamese TTS-modelle soos XTTS-v2. viVoice word vrygestel onder die CC BY-NC-SA 4.0-lisensie.

The preprocessing pipeline was designed to eliminate variability that could degrade speaker similarity. Steps included:(i) manual verification of transcript accuracy to ensure perfect alignment;(ii) text normalization using VietNormalizer(Do, 2026) to expand abbreviations, convert numbers to words, and standardize punctuation;(iii) phonetic alignment with a forced aligner trained on Vietnamese, yielding frame-level phoneme boundaries;(iv) energy-based silence stripping at the beginning and end of utterances to remove leading/trailing silence; and(v) loudness normalization to−23 LUFS to maintain consistent perceived volume across speakers. Crucially, for each speaker a high-quality speaker embedding was extracted using the Resemblyzer encoder and stored as a reference vector. Conditioning the generative model on these embeddings during training encourages the decoder to preserve speaker identity, which directly contributed to the high cosine similarity scores (0.90+) observed in the experiments.
Die voorverwerkingspyplyn is ontwerp om wisselvalligheid uit te skakel wat sprekerooreenkoms kan verswak. Stappe het ingesluit: (i) handmatige verifikasie van transkripsie-akkuraatheid om perfekte belyning te verseker; (ii) teksnormalisering met behulp van VietNormalizer (Do, 2026) om afkortings uit te brei, nommers na woorde om te skakel en leestekens te standaardiseer; (iii) fonetiese belyning met 'n gedwonge belyning wat op Viëtnamees opgelei is, wat foon-gebaseerde raamwerk lewer; stilte-stroop aan die begin en einde van uitsprake om leidende/sluitende stilte te verwyder; en (v) hardheid normalisering tot -23 LUFS om konsekwente waargenome volume oor sprekers te handhaaf. Van kardinale belang, vir elke spreker is 'n hoë-gehalte luidspreker inbedding onttrek met behulp van die Resemblyzer enkodeerder en gestoor as 'n verwysing vektor. Die kondisionering van die generatiewe model op hierdie inbeddings tydens opleiding moedig die dekodeerder aan om sprekeridentiteit te bewaar, wat direk bygedra het tot die hoë cosinus-ooreenkomstellings (0.90+) wat in die eksperimente waargeneem is.

Experimental Setup. The evaluation was conducted on a GIGABYTE G5 GD laptop, a gaming-oriented notebook whose specifications are listed in Table 1. Notably, the system features a dedicated NVIDIA GeForce RTX GPU(Ampere architecture, discrete) alongside an integrated Intel UHD Graphics adapter. However, the PyTorch environment used for inference was a CPU-only build, meaning all synthesis operations ran on the Intel Core i5-11400H CPU without GPU acceleration. This configuration reflects a common deployment scenario where CUDA drivers or libraries are not installed, or where the system prioritizes compatibility over maximum speed.

Eksperimentele opstelling. Die evaluering is uitgevoer op 'n GIGABYTE G5 GD-skootrekenaar, 'n speletjie-georiënteerde notaboek waarvan die spesifikasies in Tabel 1 gelys word. Die stelsel het veral 'n toegewyde NVIDIA GeForce RTX GPU (Ampere-argitektuur, diskreet) saam met 'n geïntegreerde Intel UHD-grafiese adapter. Die PyTorch-omgewing wat vir afleidings gebruik is, was egter slegs 'n SVE-bou, wat beteken dat alle sintese-bewerkings op die Intel Core i5-11400H SVE gehardloop het sonder GPU-versnelling. Hierdie opstelling weerspieël 'n algemene ontplooiingscenario waar CUDA-bestuurders of biblioteke nie geïnstalleer is nie, of waar die stelsel versoenbaarheid bo maksimum spoed prioritiseer.

Table 1. System configuration of the test platform.
Tabel 1. Stelselkonfigurasie van die toetsplatform.

Component Komponent	Specification Spesifikasie
Model	GIGABYTE G5 GD
CPU	Intel Core i5-11400H (8 cores, 12 threads, 2.70 GHz base)
RAM	16 GB DDR4 (15.8 GB usable)
GPU	Intel UHD Graphics (integrated)
Discrete GPU	NVIDIA GeForce RTX (Ampere, dedicated)
OS	Windows 11 Home Single Language
Model	GIGABYTE G5 GD
SVE	Intel Core i5-11400H (8 kerne, 12 drade, 2,70 GHz basis)
RAM	16 GB DDR4 (15,8 GB bruikbaar)
GPU	Intel UHD-grafika (geïntegreer)
Diskrete	NVIDIA GeForce RTX (Ampere, toegewyd)
GPU-	Windows 11 Huis Enkeltaal
bedryfstelsel	

(Build 26200)
(Bou 26200)

Inference Backend	PyTorch CPU-only, 4 threads
Afleiding Backend	PyTorch Slegs SVE, 4 drade

Voice Similarity and Speech Quality The system was evaluated by comparing original and synthesized(cloned) speech samples from four speakers. Each sample pair consisted of a reference wav file and its clone. Two assessment methods were applied: Resemblyzer for voice similarity (Wan et al., 2018; Mishra et al., 2023), and DNSMOS for perceived audio quality(Reddy et al., 2021). Voice similarity results are presented in Table 2. All pairs achieve a cosine similarity of approximately 0.90 (ranging from 0.9013 to 0.9346), confirming that the VITS model, despite running on the CPU, preserves vocal timbre and speaker identity remarkably well(Jia et al., 2018).
Stemooreenkoms en spraakkwaliteitDie stelsel is geëvalueer deur oorspronklike en gesintetiseerde (gekloonde) spraakmonsters van vier sprekers te vergelyk. Elke monsterpaar het bestaan uit 'n verwysing wav-lêer en sy kloon. Twee assesseringsmetodes is toegepas: Resemblyzer vir stemooreenkoms (Wan et al., 2018; Mishra et al., 2023), en DNSMOS vir waargenome klankkwaliteit (Reddy et al., 2021). Stemooreenkomsresultate word in Tabel 2 aangebied. Alle pare bereik 'n cosinus-ooreenkoms van ongeveer 0.90 (wat wissel van 0.9013 tot 0.9346), wat bevestig dat die VITS-model, ten spyte van die gebruik van die SVE, vokale timbre en sprekeridentiteit merkwaardig goed bewaar (Jia et al., 2018).

Voice similarity results for the viXTTS model are presented in Table 2. All pairs achieve a cosine similarity of approximately 0.90(ranging from 0.9013 to 0.9346), confirming that the model preserves vocal timbre and speaker identity remarkably well(Jia et al., 2018). Speaker similarity for other engines was not measured in this evaluation run, as most of them do not natively support voice cloning or require a separate reference embedding extraction step.
Stemooreenkoms-resultate vir die viXTTS-model word in Tabel 2 aangebied. Alle pare bereik 'n cosinus-ooreenkoms van ongeveer 0.90 (wat wissel van 0.9013 tot 0.9346), wat bevestig dat die model vokale timbre en sprekeridentiteit merkwaardig goed bewaar (Jia et al., 2018). Luidspreker-ooreenkoms vir ander enjins is nie in hierdie evaluasielopie gemeet nie, aangesien die meeste van hulle nie inheems stemkloning ondersteun nie of 'n aparte verwysing-inbedding-onttrekkingstap vereis.

Table 2. Voice similarity between original and cloned speech.
Tabel 2. Stemooreenkoms tussen oorspronklike en gekloonde spraak.

Cặp audio Knap klank Lekker klank	Similarity Ooreenkomstigheid
giong_goc và giong_clone	0.9038
Danh_1 và Danh_clone	0.9019
Khang1 và Khang_clone	0.9346
Ngoc1 và Ngoc_clone	0.9013

giong_goc en giong_clone	0,9038
Danh_1 en Danh_clone	0,9019
Khang1 en Khang_clone	0,9346
Ngoc1 en Ngoc_clone	0,9013
	0,9038
	0,9019
	0,9346
	0,9013
	0,9038
	0,9019
	0,9346
	0,9013

Speech quality results from DNSMOS across all seven evaluated engines are summarized in Table 3. Four metrics were recorded: OVRL(overall quality), SIG (speech quality), BAK (background noise intrusiveness), and the supplementary P808 score (an overall listening quality estimator aligned with ITU-T P.808). Scores are on the standard 1–5 Mean Opinion Score scale, where higher values indicate better perceptual quality.

Spraakkwaliteitresultate van DNSMOS oor al sewe geëvalueerde enjins word in Tabel 3 opgesom. Vier maatstawwe is aangeteken: OVRL (algehele kwaliteit), SIG (spraakkwaliteit), BAK (agtergrondgeraas-indringendheid), en die aanvullende P808-telling ('n algehele luisterkwaliteitberamer wat met ITU-T P.808 belyn is). Tellings is op die standaard 1–5 Gemiddelde Opinietellingskaal, waar hoër waardes beter perseptuele kwaliteit aandui.

Table 3. DNSMOS evaluation of synthesized Vietnamese speech quality..
Tabel 3. DNSMOS-evaluering van gesintetiseerde Viëtnamese spraakkwaliteit..

Model	File audio Lêer oudio	OVRL MOS	SIG MOS	BAK MOS	P808 MOS
Proposed viXTTS-based Voorgestelde viXTTS-gebaseer	giong_clone.wav	3.38	3.60	4.14	4.03
		3,38	3,60		
Voice Cloning System Stemkloningstelsel					
Piper Vietnamese TTS	Piper_Vietnamese_TTS	3.00	3.40	3.81	3.15
	Piper_Viëtnamees_TTS		3,40	3,81	
Valtec Vietnamese TTS	Valtec Vietnamese	3.07	3.34	4.04	3.95
Valtec Viëtnamese TTS	Valtec Viëtnamees				3,95
v2	TTS_ver2.wav		3,34		
VietnameseTTS	VietnameseTTS.wav	2.07	2.75	2.86	2.73
Viëtnamese TTS	ViëtnameesTTS.wav		2,75	2,86	2,73
Valtec Vietnamese TTS				4.16	4.21
Valtec Viëtnamese TTS	Valtec_Vietnamese.wav	3.37	3.58		
	Valtec_Viëtnamese.wav	3,37	3,58		
VietTTS				4.15	3.64
	VietTTS.wav	3.05	3.25		3,64
			3,25		
VieNeu-TTS	VietNeu_TTS.wav	3.31	3.54	4.16	3.68
			3,54		3,68

The DNSMOS evaluation results show noticeable differences in synthesized Vietnamese speech quality among the evaluated TTS systems. Overall, the proposed viXTTS-based voice cloning system, represented by giong_clone.wav, achieves the highest OVRL MOS score of 3.38 and the highest SIG MOS score of 3.60. These results indicate that the proposed system produces speech with strong overall perceptual quality and clear speech signals. The high BAK MOS score of 4.14 also suggests that the generated audio contains very little background noise. Therefore, the viXTTS-based model demonstrates strong performance in both speech clarity and noise suppression.

Die DNSMOS-evalueringsresultate toon merkbare verskille in gesintetiseerde Viëtnamese spraakkwaliteit tussen die geëvalueerde TTS-stelsels. Oor die algemeen behaal die voorgestelde viXTTS-gebaseerde stemkloningstelsel, verteenwoordig deur giong_clone.wav, die hoogste OVRL MOS-telling van 3.38 en die hoogste SIG MOS-telling van 3.60. Hierdie resultate dui aan dat die voorgestelde stelsel spraak produseer met sterk algehele perseptuele kwaliteit en duidelike spraakseine. Die hoë BAK MOS-telling van 4.14 dui ook daarop dat die gegenereerde klank baie min agtergrondgeraas bevat. Daarom demonstreer die viXTTS-gebaseerde model sterk prestasie in beide spraakhelderheid en geraasonderdrukking.

Among the compared systems, Valtec_Vietnamese.wav also performs competitively, with an OVRL MOS of 3.37 and a SIG MOS of 3.58, which are very close to the proposed system. In addition, this model obtains the highest P808 MOS score of 4.21, suggesting that listeners may perceive its overall audio quality favorably. However, the proposed giong_clone.wav still slightly outperforms it in the two main DNSMOS indicators, OVRL and SIG. This suggests that the proposed model has a small but measurable advantage in terms of overall speech quality and signal clarity.

Onder die vergelykende stelsels presteer Valtec_Vietnamese.wav ook mededingend, met 'n OVRL MOS van 3.37 en 'n SIG MOS van 3.58, wat baie na aan die voorgestelde stelsel is. Boonop behaal hierdie model die hoogste P808 MOS-telling van 4.21, wat daarop dui dat luisteraars die algehele klankgehalte daarvan gunstig kan waarneem. Die voorgestelde giong_clone.wav presteer egter steeds effens beter as dit in die twee hoof DNSMOS-aanwysers, OVRL en SIG. Dit dui daarop dat die voorgestelde model 'n klein maar meetbare voordeel het in terme van algehele spraakkwaliteit en seinhelderheid.

The VietNeu_TTS.wav model also achieves strong results, with an OVRL MOS of 3.31, SIG MOS of 3.54, BAK MOS of 4.16, and P808 MOS of 3.68. These scores place it in the high-performing group, especially in terms of background noise suppression. Its BAK score is among the highest in the comparison, showing that the generated audio is clean and stable. Although its overall and signal quality scores are slightly lower than those of giong_clone.wav and Valtec_Vietnamese.wav, it remains a reliable model for Vietnamese speech synthesis.

Die VietNeu_TTS.wav-model behaal ook sterk resultate, met 'n OVRL MOS van 3.31, SIG MOS van 3.54, BAK MOS van 4.16 en P808 MOS van 3.68. Hierdie tellings plaas dit in die hoëpresterende groep, veral in terme van agtergrondgeraasonderdrukking. Die BAK-telling daarvan is van die hoogste in die vergelyking, wat wys dat die gegenereerde klank skoon en stabiel is. Alhoewel sy algehele en seinkwaliteittellings effens laer is as dié van giong_clone.wav en Valtec_Vietnamese.wav, bly dit 'n betroubare model vir Viëtnamese spraaksintese.

The middle-performing group includes Valtec Vietnamese TTS_ver2.wav, VietTTS.wav, and Piper_Vietnamese_TTS. These models obtain OVRL MOS scores ranging from 3.00 to 3.07, indicating acceptable but not outstanding synthesized speech quality. Their SIG MOS scores, ranging from 3.25 to 3.40, suggest that the speech signal is reasonably clear but still less natural or less detailed than the top performing systems. Notably, VietTTS.wav achieves a high BAK MOS score of 4.15, showing that its audio output is relatively clean. However, its lower OVRL and SIG scores indicate that background cleanliness alone is not sufficient to ensure high overall speech quality.
Die middelpresterende groep sluit Valtec Vietnamese TTS_ver2.wav, VietTTS.wav en Piper_Vietnamese_TTS in. Hierdie modelle verkry OVRL MOS-tellings wat wissel van 3.00 tot 3.07, wat aanvaarbare maar nie uitstaande gesintetiseerde spraakkwaliteit aandui nie. Hul SIG MOS-tellings, wat wissel van 3.25 tot 3.40, dui daarop dat die spraaksein redelik duidelik is, maar steeds minder natuurlik of minder gedetailleerd as die toppresterende stelsels. Veral, VietTTS.wav behaal 'n hoë BAK MOS-telling van 4.15, wat wys dat die klankuitset relatief skoon is. Die laer OVRL- en SIG-tellings dui egter daarop dat agtergrondskoonheid alleen nie voldoende is om hoë algehele spraakkwaliteit te verseker nie.

The lowest-performing system is VietnameseTTS.wav, which obtains an OVRL MOS of 2.07, SIG MOS of 2.75, BAK MOS of 2.86, and P808 MOS of 2.73. These scores are substantially lower than those of the other models. In particular, the low OVRL and SIG scores suggest that the synthesized speech may contain noticeable artifacts, reduced naturalness, or unclear pronunciation. The low BAK score also indicates that the audio may include more background interference or noise compared with the other systems.
Die stelsel wat die laagste presteer is VietnameseTTS.wav, wat 'n OVRL MOS van 2.07, SIG MOS van 2.75, BAK MOS van 2.86 en P808 MOS van 2.73 verkry. Hierdie tellings is aansienlik laer as dié van die ander modelle. Die lae OVRL- en SIG-tellings dui veral daarop dat die gesintetiseerde spraak merkbare artefakte, verminderde natuurlikheid of onduidelike uitspraak kan bevat. Die lae BAK-telling dui ook aan dat die oudio meer agtergrondinterferensie of geraas kan insluit in vergelyking met die ander stelsels.

A general trend can be observed across the stronger models: most systems achieve high BAK MOS scores above 4.0, which means that modern Vietnamese TTS models are generally effective at producing clean audio with limited background noise. However, differences in OVRL and SIG scores show that clean audio does not always
'n Algemene neiging kan oor die sterker modelle waargeneem word: die meeste stelsels behaal hoë BAK MOS-tellings bo 4.0, wat beteken dat moderne Viëtnamese TTS-modelle oor die algemeen effektief is om skoon oudio met beperkte agtergrondgeraas te produseer. Verskille in OVRL- en SIG-tellings toon egter dat skoon klank nie altyd nie

guarantee natural and intelligible speech. Speech naturalness, clarity, and signal quality remain important factors in distinguishing the best-performing models.
waarborg natuurlike en verstaanbare spraak. Spraaknatuurlikheid, duidelikheid en seinkwaliteit bly belangrike faktore om die beste presterende modelle te onderskei.

In summary, the proposed viXTTS-based voice cloning model giong_clone.wav achieves the best overall performance in terms of OVRL and SIG MOS, making it the strongest system in this comparison. Valtec_Vietnamese.wav and VietNeu_TTS.wav also demonstrate competitive performance and can be considered strong alternatives. Meanwhile, VietnameseTTS.wav shows the weakest results and may require further optimization to improve speech naturalness, clarity, and background noise control.
Samevattend, die voorgestelde viXTTS-gebaseerde stemkloningmodel giong_clone.wav behaal die beste algehele prestasie in terme van OVRL en SIG MOS, wat dit die sterkste stelsel in hierdie vergelyking maak. Valtec_Vietnamese.wav en VietNeu_TTS.wav demonstreer ook mededingende prestasie en kan as sterk alternatiewe beskou word. Intussen toon VietnameseTTS.wav die swakste resultate en kan verdere optimalisering vereis om spraaknatuurlikheid, helderheid en agtergrondgeraasbeheer te verbeter.

Real-Time Performance and Latency Analysis: The most critical performance metric of the current prototype is the end-to-end synthesis latency. For a typical input text of 750 characters (equivalent to approximately 50 seconds of generated audio at a standard Vietnamese speaking rate of about 900 characters per minute), the system requires roughly 2 minutes and 50 seconds(170 seconds) to produce the final WAV file. This yields a Real-Time Factor(RTF) of approximately 3.4, well above the real-time threshold of 1.0. The latency is not fixed, however; it varies measurably depending on the linguistic complexity of the input text. Two factors dominate the additional processing overhead: Emotion tags. Each emotional boundary forces a new synthesis chunk with a separate viXTTS inference call, leading to multiple context switch and model loading overheads. More tags increase the number of chunks and the cumulative pause insertion time.
Intydse prestasie- en vertragingsanalise: Die mees kritieke prestasiemaatstaf van die huidige prototipe is die einde-tot-einde sintese latency. Vir 'n tipiese invoerteks van 750 karakters (gelykstaande aan ongeveer 50 sekondes se gegenereerde oudio teen 'n standaard Viëtnamees-sprekende tempo van ongeveer 900 karakters per minuut), benodig die stelsel ongeveer 2 minute en 50 sekondes (170 sekondes) om die finale WAV-lêer te produseer. Dit lewer 'n Real-Time Factor (RTF) van ongeveer 3.4, ver bo die intydse drempel van 1.0. Die latensie is egter nie vas nie; dit wissel meetbaar na gelang van die linguistiese kompleksiteit van die invoerteks. Twee faktore oorheers die bykomende verwerkingsbokoste: Emosie-etikette. Elke emosionele grens dwing 'n nuwe sintese-stuk af met 'n aparte viXTTS-afleidingsoproep, wat lei tot veelvuldige konteksskakelaar en modellaai bokoste. Meer merkers verhoog die aantal stukke en die kumulatiewe pouse-invoegtyd.

English words and non Vietnamese segments. These require extra grapheme to phoneme steps within the text normalization layer (Do, 2026), and may introduce attention errors that force the decoder to retry alignment, adding to the total runtime.
Engelse woorde en nie-Viëtnamese segmente. Dit vereis ekstra grafeem-tot-foneemstappe binne die teksnormaliseringslaag (Do, 2026), en kan aandagfoute

veroorsaak wat die dekodeerder dwing om herbelyning te probeer, wat bydra tot die totale looptyd.

The observed latency stems from running the full VITS pipeline on the CPU without utilizing the available RTX GPU. The CPU-only PyTorch build cannot exploit the massive parallelism of the GPU's tensor cores, leaving matrix multiplications and convolutions to be computed sequentially. Despite this, the system remains acceptable for near-line content creation tasks such as podcast dubbing, educational narration, or voicemail generation—applications where a few minutes of rendering time are tolerated in exchange for high-quality, emotionally expressive output.
Die waargenome latensie spruit uit die loop van die volle VITS-pyplyn op die SVE sonder om die beskikbare RTX GPU te gebruik. Die SVE-enigste PyTorch-bou kan nie die massiewe parallelisme van die GPU se tensorkerne ontgin nie, wat matriksvermenigvuldigings en konvolusies laat om opeenvolgend bereken te word. Ten spyte hiervan, bly die stelsel aanvaarbaar vir nabylyn-inhoudskeppingstake soos podcast-oorklanking, opvoedkundige vertelling of stemposgenerering – toepassings waar 'n paar minute se leweringstyd geduld word in ruil vir hoëgehalte, emosioneel ekspressiewe uitset.

Critically, the machine is equipped with an NVIDIA GeForce RTX GPU, which is fully capable of accelerating the inference. Moving to a CUDA-enabled PyTorch installation would reduce the RTF to well below 0.1 (based on typical GPU benchmarks for VITS), enabling real-time streaming synthesis. In addition, model quantization to FP16 or INT8, and switching to optimized inference engines like ONNX Runtime with OpenVINO, could further halve latency while preserving the high similarity and MOS scores.
Die masjien is krities toegerus met 'n NVIDIA GeForce RTX GPU, wat ten volle in staat is om die afleiding te versnel. Om na 'n CUDA-geaktiveerde PyTorch-installasie te skuif, sal die RTF tot ver onder 0.1 verminder (gebaseer op tipiese GPU-maatstawwe vir VITS), wat intydse stroomsintese moontlik maak. Daarbenewens kan modelkwantisering na FP16 of INT8, en oorskakeling na geoptimaliseerde inferensie-enjins soos ONNX Runtime met OpenVINO, latensie verder halveer terwyl die hoë ooreenkomste en MOS-tellings behoue bly.

6 Conclusion
6 Gevolgtrekking

In this research, a Vietnamese text-to-speech system based on artificial intelligence and deep learning was thoroughly designed, implemented, and evaluated. The system
In hierdie navorsing is 'n Viëtnamese teks-na-spraak-stelsel wat op kunsmatige intelligensie en diep leer gebaseer is, deeglik ontwerp, geïmplementeer en geëvalueer. Die stelsel

exploits modern models such as Tacotron, FastSpeech, and VITS to automatically convert text into natural speech signals, learning the unique phonetic and prosodic features of Vietnamese. Experimental outcomes show that the proposed model generates clear, natural speech with appropriate intonation and maintains stable performance across various conditions and contexts.
ontgin moderne modelle soos Tacotron, FastSpeech en VITS om teks outomaties in natuurlike spraakseine om te skakel, en leer die unieke fonetiese en prosodiese kenmerke van Viëtnamees. Eksperimentele uitkomste toon dat die voorgestelde model duidelike, natuurlike spraak met toepaslike intonasie genereer en stabiele prestasie oor verskeie toestande en kontekste handhaaf.

The achieved results not only validate the effectiveness of deep learning in Vietnamese speech synthesis but also highlight the system's broad applicability in practice, notably in virtual assistants, education, automatic call centers, and assistive tools for the visually impaired. Future work may focus on expanding and enriching the speech dataset, improving regional naturalness, optimizing models for resource-constrained devices, and integrating the system into real-world platforms to enhance feasibility and usability.
Die resultate wat behaal is, bevestig nie net die doeltreffendheid van diep leer in Viëtnamese spraaksintese nie, maar beklemtoon ook die stelsel se breë toepaslikheid in die praktyk, veral in virtuele assistente, onderwys, outomatiese oproepsentrums en hulpmiddels vir gesiggestremdes. Toekomstige werk kan fokus op die uitbreiding en verryking van die spraakdatastel, die verbetering van plaaslike natuurlikheid, die optimalisering van modelle vir hulpbronbeperkte toestelle, en die integrasie van die stelsel in werklike wêreldplatforms om uitvoerbaarheid en bruikbaarheid te verbeter.

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